Adversarial Style Transfer for Bengali Text

Domain-Adversarial Neural Network with GRU Encoder–Decoder

Group: Alchemists

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Motivation — Why Style Transfer?

△ The Problem

- Style transfer **well-studied in English**, almost **absent in Bengali**
- Many Bengali Authors do not have digitised representation of their works
- No publicly available **sentence-author labelled dataset** exists in public repositories.

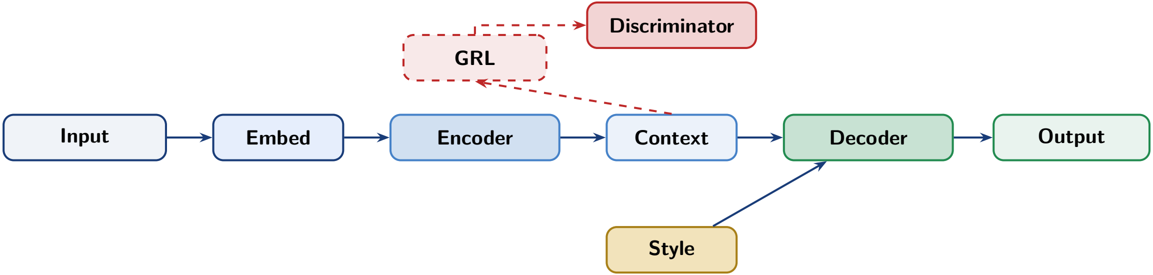
⋆ Applications

- Literary education & authorship analysis
- Creative writing assistance

Input	আজ আকাশ মেঘলা ছিল।
Tagore	আজ নীলিমা মেঘে ছাওয়া...
Sarat	আজকের আকাশখানি মেঘে ভরা...
Satyajit	আকাশে আজ মেঘ জমেছে।

Same meaning → different styles

System Architecture — Big Picture



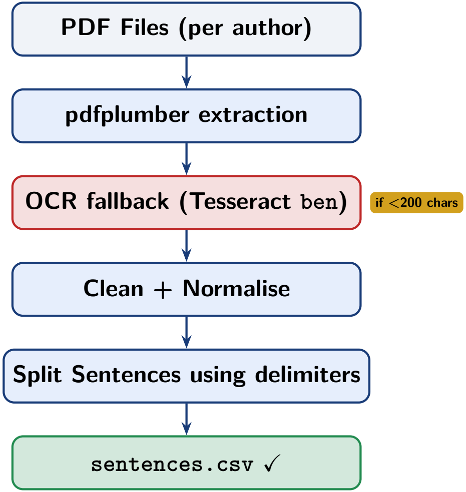
Dataset — Building the Bangla Corpus

Pipeline

1. Collect PDFs per author folder
2. Extract text via **pdfplumber**
3. Fallback: **Tesseract OCR** (Bengali model, 300 DPI)
4. Sentence validation: length, vowel density, OCR-junk filter

Corpus Statistics

Authors	**5**
Training set size	**18779**
Min word count	5 words/sentence



Dataset Overview

Author	Sent.	Noise
Tagore	3279	0.064
Sarat	3500	0.099
Satyajit	4000	0.095
Bibhuti	4500	0.116
Sunil	3500	0.084

18,779 sentences

```
sunil gangopadhay,"জনতা ভেদ করে মাঝখানে এসে গৌর গলায় বললো, মধু, তুই ফের ব্র্যাপ্ডির বোতল নিয়ে কালেজে এসেচিস"
satyajit ray,আমার আগামী উপন্যাসের তিনটি নাম্ ঠিক করা আছে কোনটা বিহর করব যদি বলে দেন
rabindranath tagore,"কিন্তু মহেন্দ্র অন্নপূর্ণা কহিলেন, না বাছা, মহেন্দ্রের সঙ্গে তাহার কোনোমতেই বিবাহ হইবার নয়"
bibhutibhushan bandopadhyay,"বিকেলের সুর্য বাগানের নিবিড় সবুজের আড়ালে ঢলে পড়েছে, এমনসময়রাজারাম রায় নিজের বাড়িতে ঢুকে ঘোড়া থেকে নামলেন"
```

Sentence–Author pairs

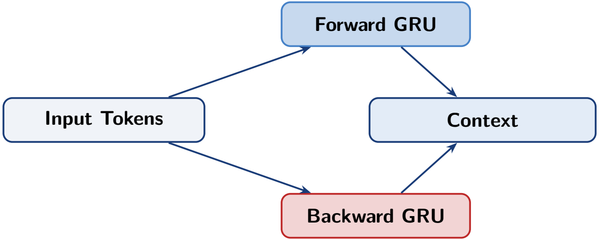
Bidirectional GRU Encoder

⊗ Design Choices

- **BiGRU**: captures left & right context simultaneously
- Forward hidden $h_{\rightarrow}$ + Backward hidden $h_{\leftarrow}$ concatenated → **260-d** style-rich representation
- Padding handled via `<pad>` index in embedding
- **No** `pack_padded_sequence` — uniform padding per batch

Dimensions

`Embed:` vocab $\times$ 128
`GRU:` input=128, hidden=130, bidir
`Context:` $B \times 260$



Adversarial Training — Gradient Reversal (DANN)

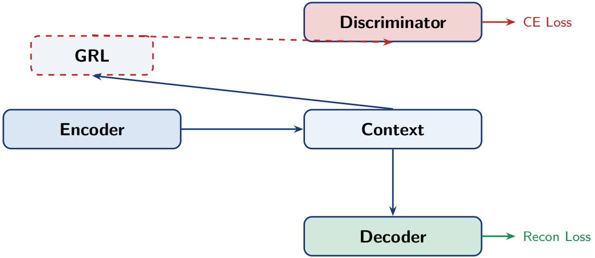
⇌ Core Idea

- **Goal:** Encoder learns representations *blind* to author identity
- **Discriminator** tries to **predict author** from encoder output
- **GRL** *negates* gradient before it reaches the encoder
- Encoder is pushed to **fool** the discriminator

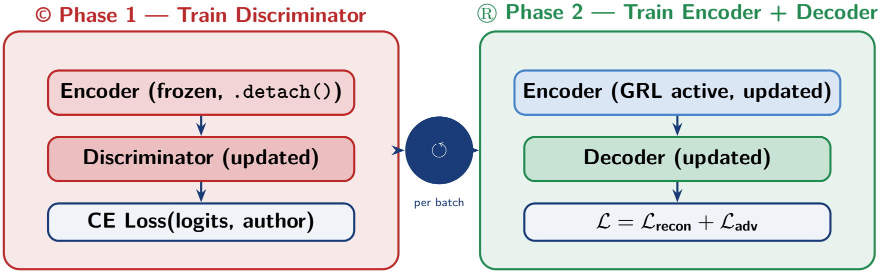
∼ λ Schedule

$$\lambda = \lambda_{\text{max}} \cdot \left(2\sigma(10p) - 1\right)$$

Warm-up: $\lambda = 0$ (first 30 epochs)



Two-Phase Adversarial Training



Save checkpoint on lowest reconstruction loss (not total loss — discriminator loss should be HIGH)

Training Setup

Hyperparameter	Value
Embedding dimension	128
Encoder GRU hidden size	130 (bidirectional $\rightarrow$ 260)
Decoder GRU hidden size	130 (unidirectional)
Author embedding dimension	128
Discriminator hidden sizes	120, 60
Batch size	64
Optimiser	Adam
Learning rate (encoder)	1×10^{-4}
Learning rate (discriminator)	1×10^{-4}
Max epochs	200
Warm-up epochs (no GRL)	30
λ_{max}	0.0001
Vocabulary min-frequency	5
Max generated length (inference)	30
Temperature	0.7
Top-k	20
Repetition penalty	1.5

Key training configuration

Training Dynamics

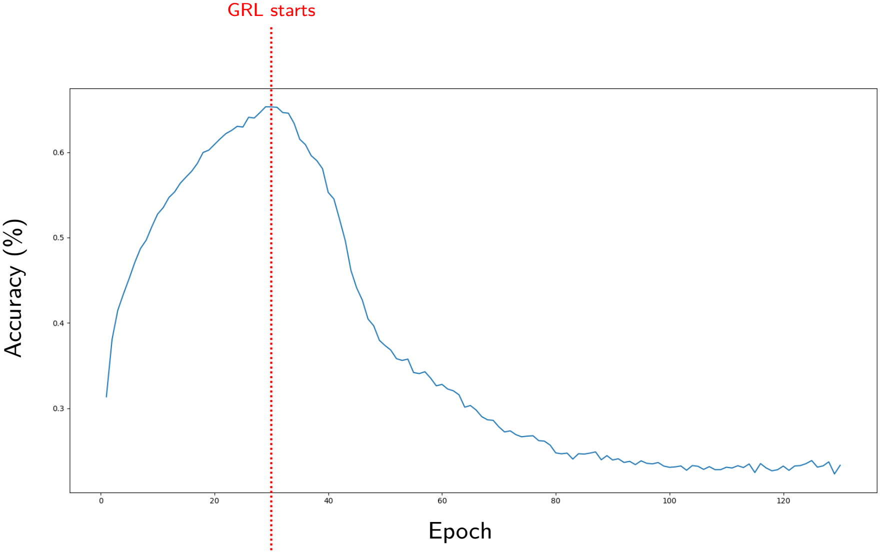
$\nearrow$ What to look for

- Accuracy rises initially (discriminator learns)
- Peaks near GRL start ($\sim$ epoch 30)
- Then drops with oscillations (adversarial learning)
- Converges near chance level ($\approx 20\%$)

★ Checkpointing

- Select model with best content + style tradeoff

Training Dynamics (Discriminator Accuracy)



Style-Conditioned Decoder

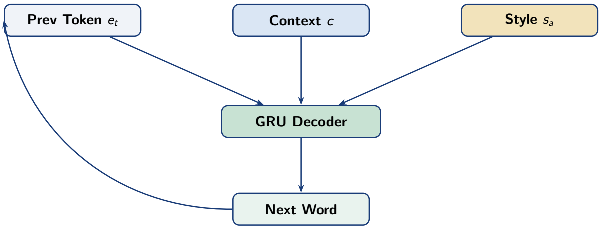
Teacher Forcing

- During training: feed **ground-truth** previous token
- Avoids compounding errors during training
- At inference: autoregressive (**top-k** + **temperature**)

Input to Each Decoder Step

$$\mathbf{d}_t = [\mathbf{e}_t \parallel \mathbf{c} \parallel \mathbf{s}_a]$$

$\mathbf{e}_t$ word embed (128-d)
$\mathbf{c}$ encoder context (style-invariant) (260-d)
$\mathbf{s}_a$ author style (128-d)
Total input = **516-d**



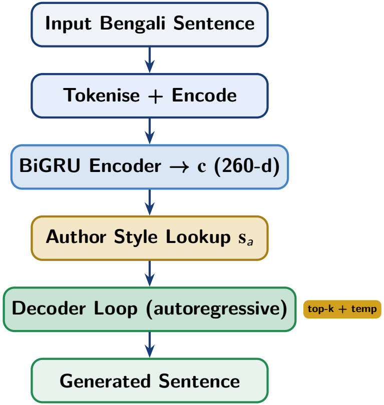
Inference — Autoregressive Generation

◇ Generation Tricks

- **Temperature** $T = 0.7$ — sharper distribution, less chaos
- **Top-k filtering** $k = 20$ — eliminates long-tail noise
- **Repetition penalty** $\rho = 1.5$ — divides logits of seen tokens
- Block: `<pad>`, `<unk>`, `<sos>`
- Stop at `<eos>` or `max_len=30`

↓ Sampling

$$p_i = \text{softmax}\left(\frac{\ell_i/\rho_i}{T}\right), \quad \hat{w}_t \sim \text{Multinomial}(p)$$



Model Outputs (Style Transfer)

Tagore

```
1. তুই কোথা থেকে এসেছিস -------> তা তুই মাপ থেকে বন্দে হইতে  আমার
2. আপনার সাথে দেখা হয়ে ভালো লাগলো   -------> আপনার দেখা ওর পরের এই সে যে আসে রসের এ-সব দিলেন এর ছিল না
```

Sarat Chandra

```
1. সে গান গায় -----------> আমি যে গায় সে দিলেন
2. শিক্ষক গল্প বলে -----------> বলে আমাকে নিলু মনোরমা এর
```

Satyajit Ray

```
1. তুই কোথা থেকে এসেছিস --------> তা হলে দেখলাম তুই আমার শখের যে
2. আপনি কি বোঝেন --------> এর কি আপনি ছিল আমার মগনলাল
```

Bibhutibhushan

```
1. বাড়িতে আমি আছি --------> আমি কোন আমার স্বামীর কথাই বলি
2. সে গান গায় ----------->  সে দেশের অজানা যে গান ছিল
```

Sunil Gangopadhyay

```
1. আপনি কি বোঝেন --------> আমার তো আপনি কি আচে না,  ঠিক
2. অনেক দিন দেখা হয়না ---------> আমার রাত্রে তিনজন কদিন অবশ্য ফিতে ছিল
3. বাড়িতে আমি আছি   ---------> আমি বললুম,  আমার যেই যে সে বাড়িতে
```

Same content $\rightarrow$ different author styles

Challenges & Limitations

△ Challenges Faced

- **OCR noise**: Bengali script complex; pdfplumber often fails on scanned books
- **No parallel corpus**: cannot compute true BLEU (style-parallel reference unavailable)
- λ **tuning**: too high $\rightarrow$ accuracy crash; too low $\rightarrow$ no style disentanglement
- **Out-of-vocabulary**: rare/dialectal words defaulted to `<unk>`
- **Long sentences**: uniform padding inflates memory usage

▶ Future Directions

- Replace GRU with **LSTM or Transformer** encoder-decoder
- Use **BanglaBERT** embeddings for richer representations
- Make the Dataset creation pipeline **more efficient** (To increase dataset size without taking long time)
- **Beam search** at inference for higher quality

Conclusion

✓ What We Built

1. Custom **Bengali literary corpus** builder (OCR pipeline)
2. **BiGRU encoder** with adversarial style disentanglement
3. **Gradient Reversal Layer** — author-blind representations
4. Style-conditioned **UniGRU decoder** with teacher forcing
5. **Inference pipeline** with temperature + top-k + rep. penalty

References I

Shen et al. (2017). *Style Transfer from Non-Parallel Text by Cross-Alignment*. NeurIPS.

Hu et al. (2017). *Toward Controlled Generation of Text*. ICML.

Prabhumoye et al. (2018). *Style Transfer Through Back-Translation*. ACL.

Dai et al. (2019). *Style Transformer: Unpaired Text Style Transfer*. ACL.

Ganin et al. (2016). *Domain-Adversarial Training of Neural Networks*. JMLR.

PRESENTATION COMMENTS

✓ Comments Provided From Presentation

1. Utilize Sub-word Tokenization as a replacement of the current tokenization strategy. - **Ongoing (To be submistted by Monday)**
2. Keep the Dataset generated in a public repository. - **DONE - Dataset is available here:** **`https://github.com/csagnik1302/DL-proj/blob/main/Dataset_Creator/bangla_corpus/sentences.csv`**
3. Use pypdf reader as a replacement of Tesseract OCR for reading PDF files which are not text-enabled for improving efficiency of the dataset generation pipeline - **Pending**